Supplementary information of

A skin-conformal rigid-in-soft array-based imaging system

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Supplementary Figures

Supplementary Notes

Supplementary Figures

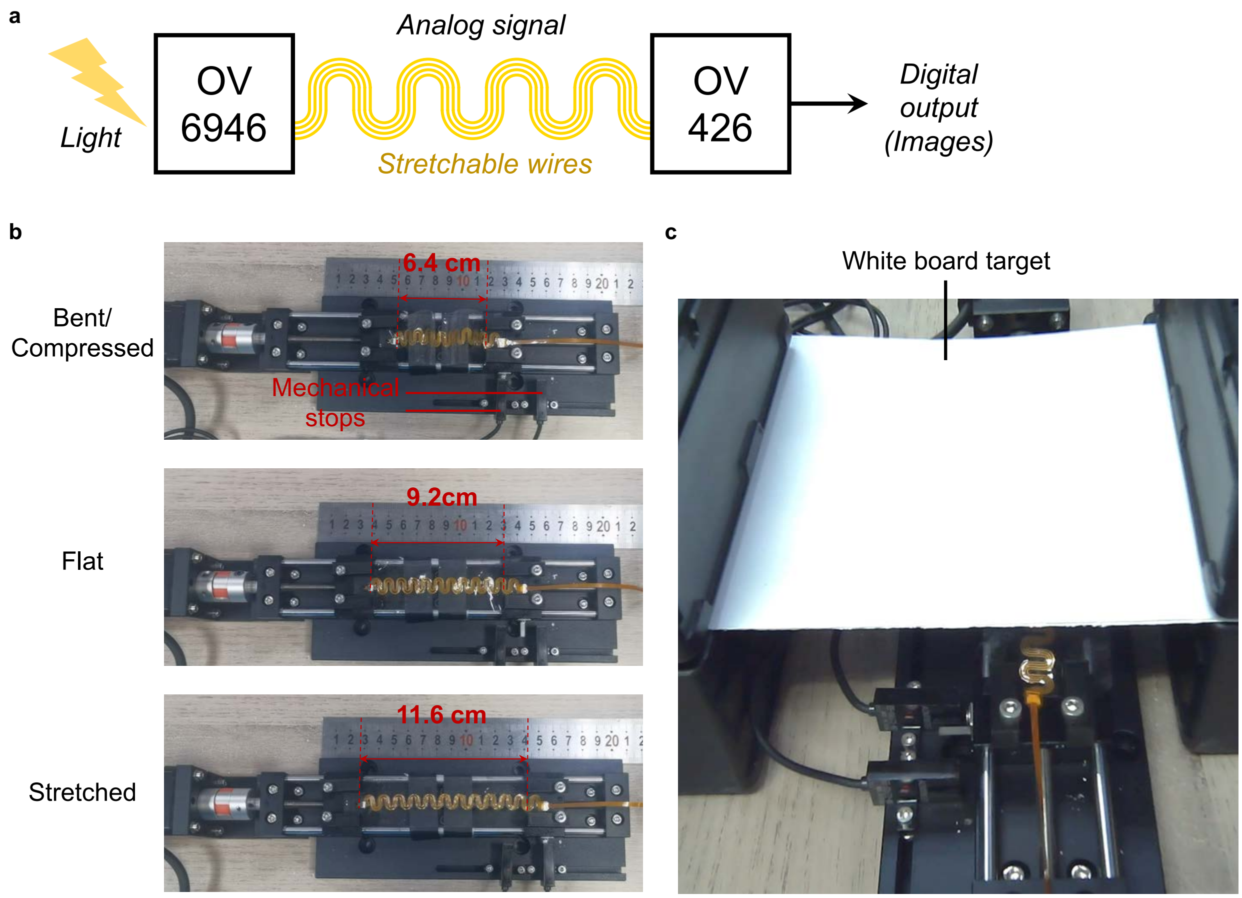
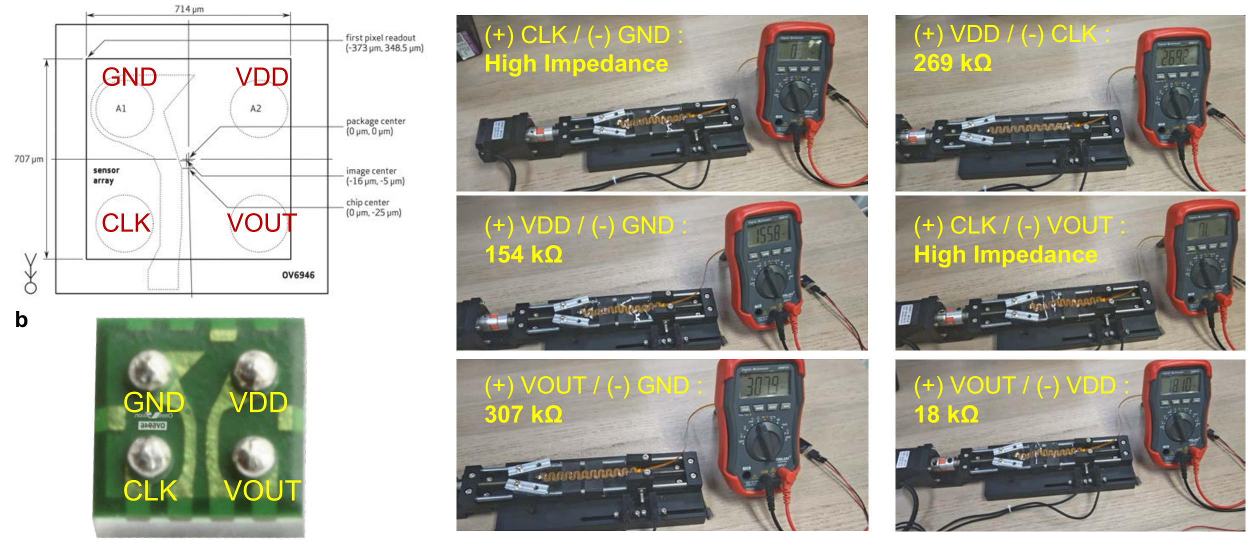


Fig. S1: Experimental setup for cyclic evaluations (see Movie S1). a, Signal transmission architecture of the SkinSight system. The stretchable interconnects transmit proprietary encoded analog signals from the miniaturized CMOS imager (OV6946) to the bridge processor (OV426). The OV426 subsequently decodes and digitizes the input into standard video outputs. This active encoding-decoding scheme inherently resists electrical noise. **b,** Cycle loading test platform. The SkinSight system is securely clamped onto a motorized linear stage, undergoing continuous reciprocating motion across three defined states: compressed (6.4 cm), flat (9.2 cm), and stretched (11.6 cm). **c,** A uniform white board is positioned as the imaging target to evaluate the dynamic optical performance during the cyclic testing.



d

Test Points (+ / -)	(+) CLK / (-) GND	(+) VDD / (-) GND	(+) VOUT / (-) GND	(+) VDD / (-) CLK	(+) CLK / (-) VOUT	(+) VOUT / (-) VDD
Measured Resistance	High Impedance	153.5~155.4 kΩ	306.1~309 kΩ	268.4~270.0 kΩ	High Impedance	18.05~18.27 kΩ

Fig. S2: Electrical resistance stability at the rigid-flex interface (see Movie S2). a,b, The pinout configuration and a photograph of the rigid sensor module. **c,d,** Continuous pin-to-pin electrical resistance measurement (e.g., VDD, GND, VOUT, VDD) using a standard digital multimeter during mechanical cyclic loading. The highly stable resistance values confirm the structural robustness of the reinforced solder joints under deformation.

Fig. S3: Quantitative evaluation of absolute metric recovery. Top: 3D bounding boxes applied to the reconstructed geometry to computationally extract the dimensions of diverse objects in a complex desktop scene. Bottom: Photographs of the corresponding physical objects alongside a standard ruler, providing the ground-truth measurements for validation.

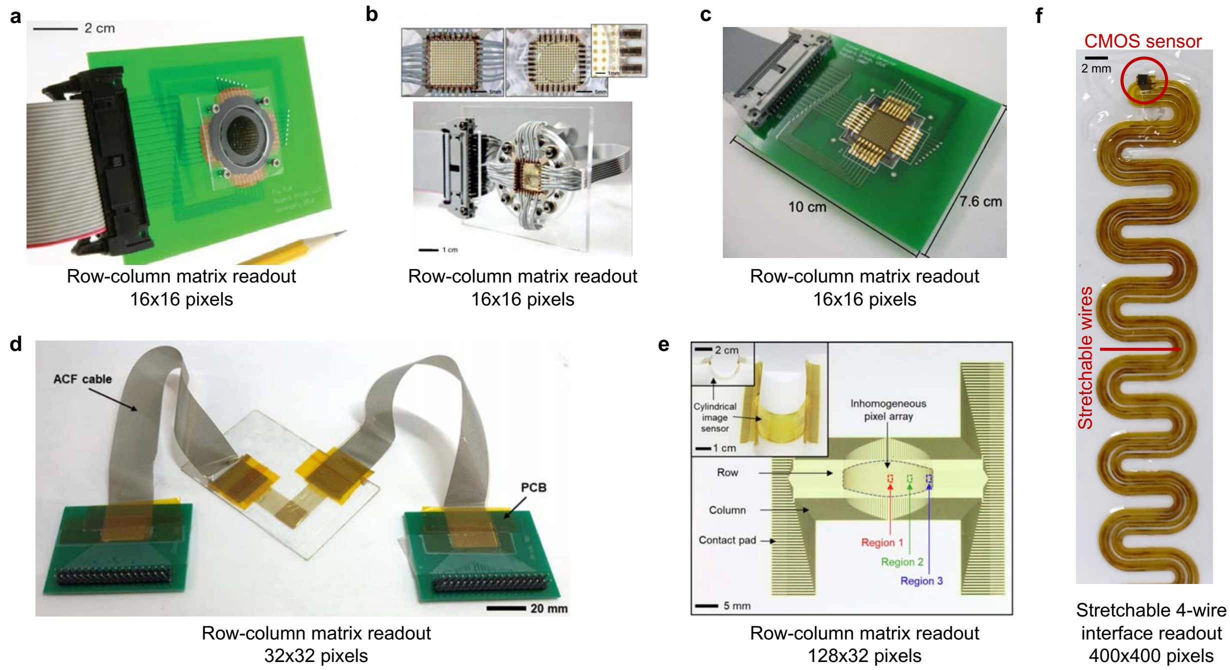


Fig. S4: Comparison of data readout and wiring architectures. a-e, Existing flexible vision systems relying on direct row and column matrix addressing. In these architectures, the number and routing complexity of the required readout interconnects increase drastically as the pixel count scales up, severely limiting the achievable resolution. **f,** The SkinSight system utilizes in-node matrix addressing (integrated in the OV6946 CMOS sensor). This rigid-in-soft architecture fundamentally decouples the pixel count from the external wiring complexity. It allows high-density image data (400x400 pixels per node) to be transmitted as encoded signals through a simple 4-wire stretchable interface, enabling exceptional modular scalability and mechanical robustness.

Fig. S5: Scalability and versatility of the SkinSight system. Representative prototypes of SkinSight configurations with varying node counts and integration scenarios, including a single sensor strip (top left), multi-node integration on wearable gloves (top middle and bottom middle), seamless palm-mounted arrays for human augmentation (bottom left), large-scale sensor sheets (bottom right), and conformal integration onto a multi-fingered soft gripper (top right).

COLMAP-sparse (40 sec.)
175 images

COLMAP-dense (20 min.)
175 images

Ours (38 sec)
9 Chunks, 4.2 sec. per chunk
175 images, 4.6 frames per sec,

Fig. S6: Comparison with traditional Structure-from-motion pipelines. Traditional SfM pipelines like COLMAP yield either overly sparse point clouds (left) or noisy, incomplete dense reconstructions (middle), whereas our approach (right) reconstructs highly robust and photorealistic 3D representations.

Fig. S7: Comparison of Gaussian reconstruction quality (InstantSplat vs. Ours). Our method (bottom) achieves superior textural fidelity and structural integrity compared to InstantSplat (top) when processing the highly deformable SkinSight video streams.

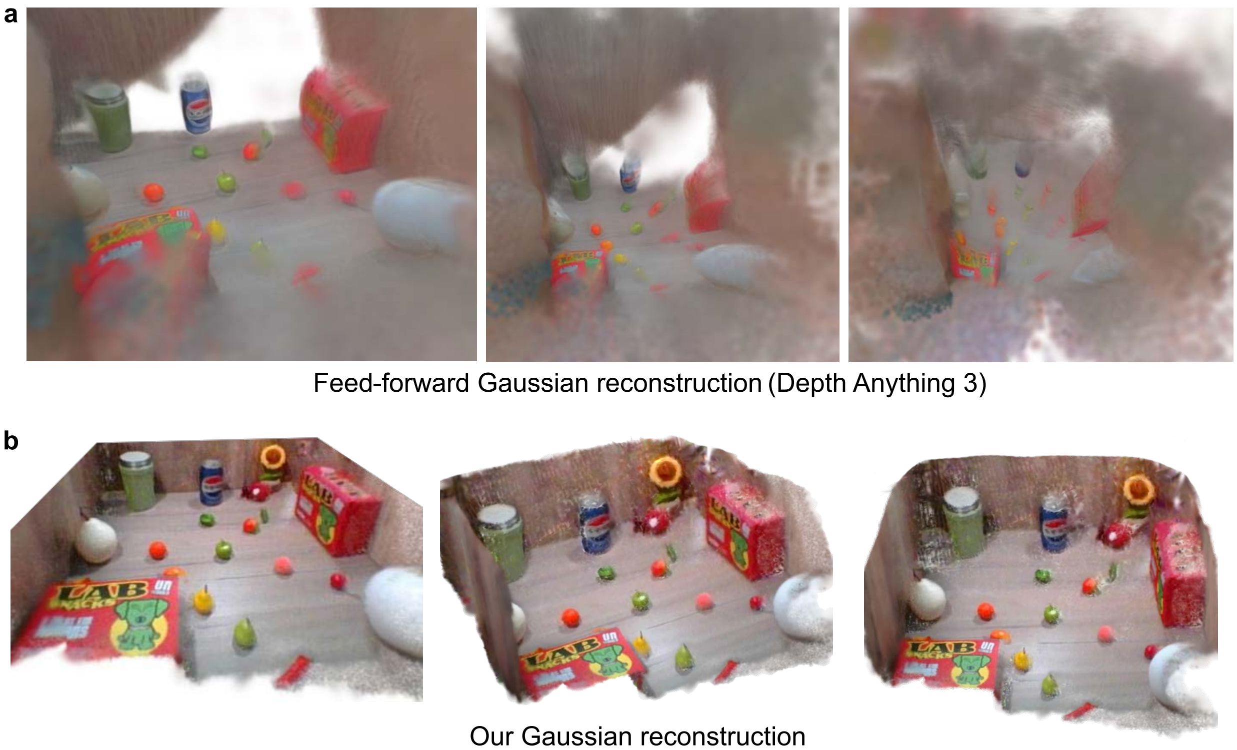
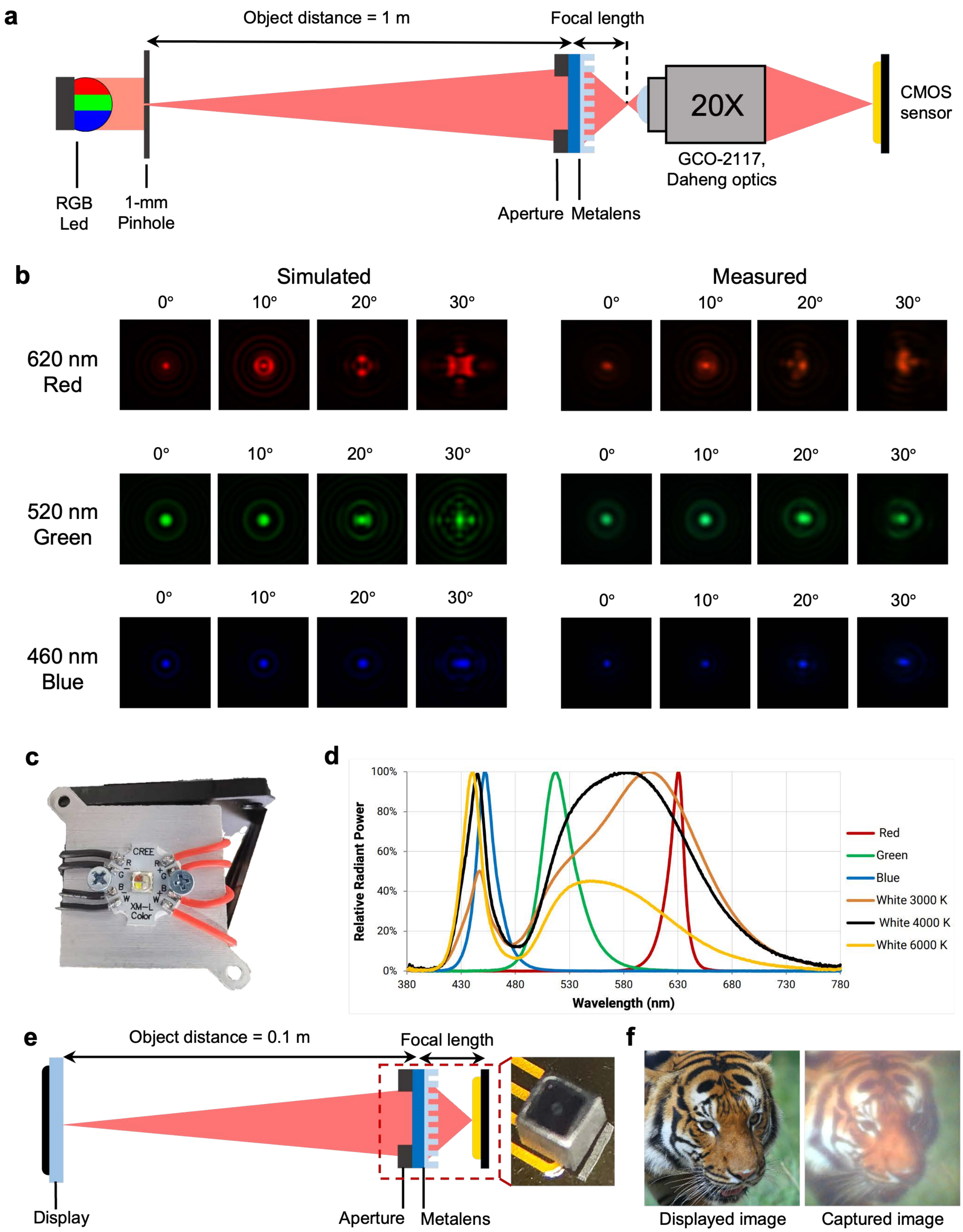


Fig. S8: Comparison of Gaussian reconstruction robustness (Feedforward Depth Anything 3 vs. Ours). Compared to feed-forward Gaussian methods (top row) which suffer from severe artifacts and structural collapse, our dual-pathway architecture (bottom row) reconstructs highly robust and photorealistic 3D representations from identical SkinSight streams.



a
Object distance = 1 m
Focal length
20X
CMOS sensor
GCO-2117, Daheng optics
RGB Led
1-mm Pinhole
Aperture
Metalens
b
Simulated
Measured
0°
10°
20°
30°
620 nm Red
520 nm Green
460 nm Blue
c
CREE
XM-L Color
d
Relative Radiant Power
100%
80%
60%
40%
20%
0%
380
430
480
530
580
630
680
730
780
Wavelength (nm)
Red
Green
Blue
White 3000 K
White 4000 K
White 6000 K
e
Object distance = 0.1 m
Focal length
Display
Aperture
Metalens
f
Displayed image
Captured image

Fig. S9: Nano-lens calibration and dataset capturing system. a, System configuration for point spread function (PSF) calibration, utilizing a color LED and a pinhole as the point light source. A 20× microscope objective lens (GCO-2117, Daheng Optics, NA = 0.4) magnifies the image to capture high-resolution PSFs. **b,** Simulated and experimentally measured PSFs under red, green, and blue point light sources at incidence angles of 0°, 10°, 20°, and 30°. Due to the circular symmetry of the nano-optics, the resulting PSFs are also symmetric. The simulated and measured PSFs show strong agreement. Note that at 30°, the PSFs extend partially beyond the NA of the microscope objective lens, resulting in incomplete capture of the PSF. **c,** Optical image of the RGBW LED (XLamp XM-L Color LEDs, Cree LED). **d,** The relative spectral power distribution of the LED. We only used the red, green, and blue dies for PSF calibration. **e,** System configuration for dataset acquisition. The ground-truth image is initially displayed on a screen and subsequently captured by our miniaturized camera. **f,** Displayed images on the screen and captured images.

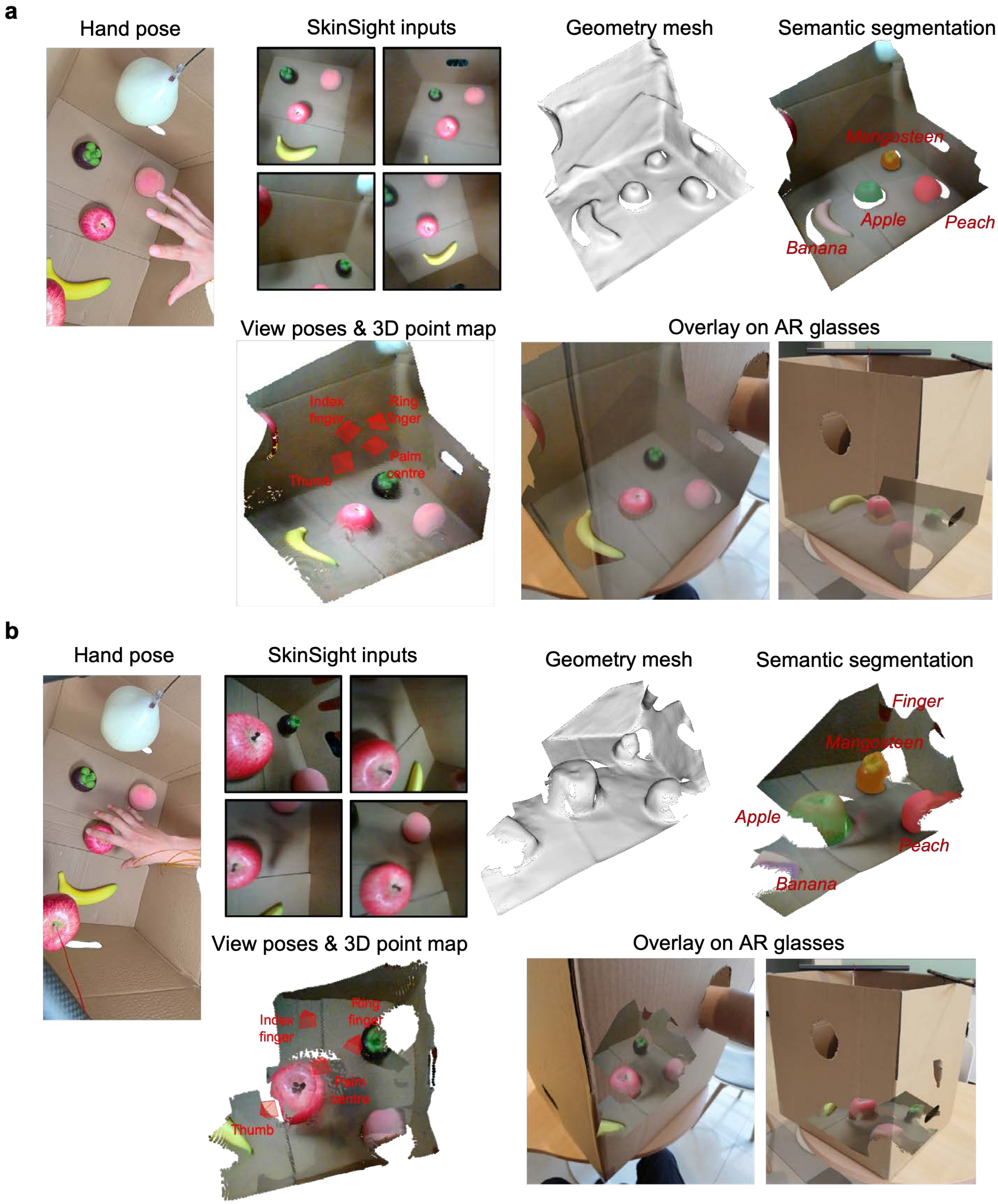

a
Hand pose
SkinSight inputs
Geometry mesh
Semantic segmentation
Mangosteen
Apple
Peach
Banana
View poses & 3D point map
Index finger
Ring finger
Palm centre
Thumb
Overlay on AR glasses
b
Hand pose
SkinSight inputs
Geometry mesh
Semantic segmentation
Finger
Mangosteen
Apple
Peach
Banana
View poses & 3D point map
Index finger
Ring finger
Palm centre
Thumb
Overlay on AR glasses

Fig. S10: Performance of SkinSight system on human hand. ab, Captured images and reconstruction results for two different hand poses. The top row shows the hand pose (optical image captured from above), the images captured by the SkinSight system, the reconstructed geometry mesh, and the semantic segmentation results. The bottom row presents the reconstructed view poses with the 3D point map, and the overlay of rendered images on AR glasses. Four miniaturized imagers positioned on the thumb, index finger, ring finger, and palm center are used to reconstruct the scene. The results demonstrate that SkinSight successfully recovers the view poses and reconstructs the geometry of both the box and the fruit objects inside. Additionally, fingertip geometry is also reconstructed, as shown in the mesh and segmentation results in b. For human applications, the reconstruction can be fed back to users via AR glasses, enabling them to "see through" objects and enhance their visual perception. See Supplementary Video 1 for more results.

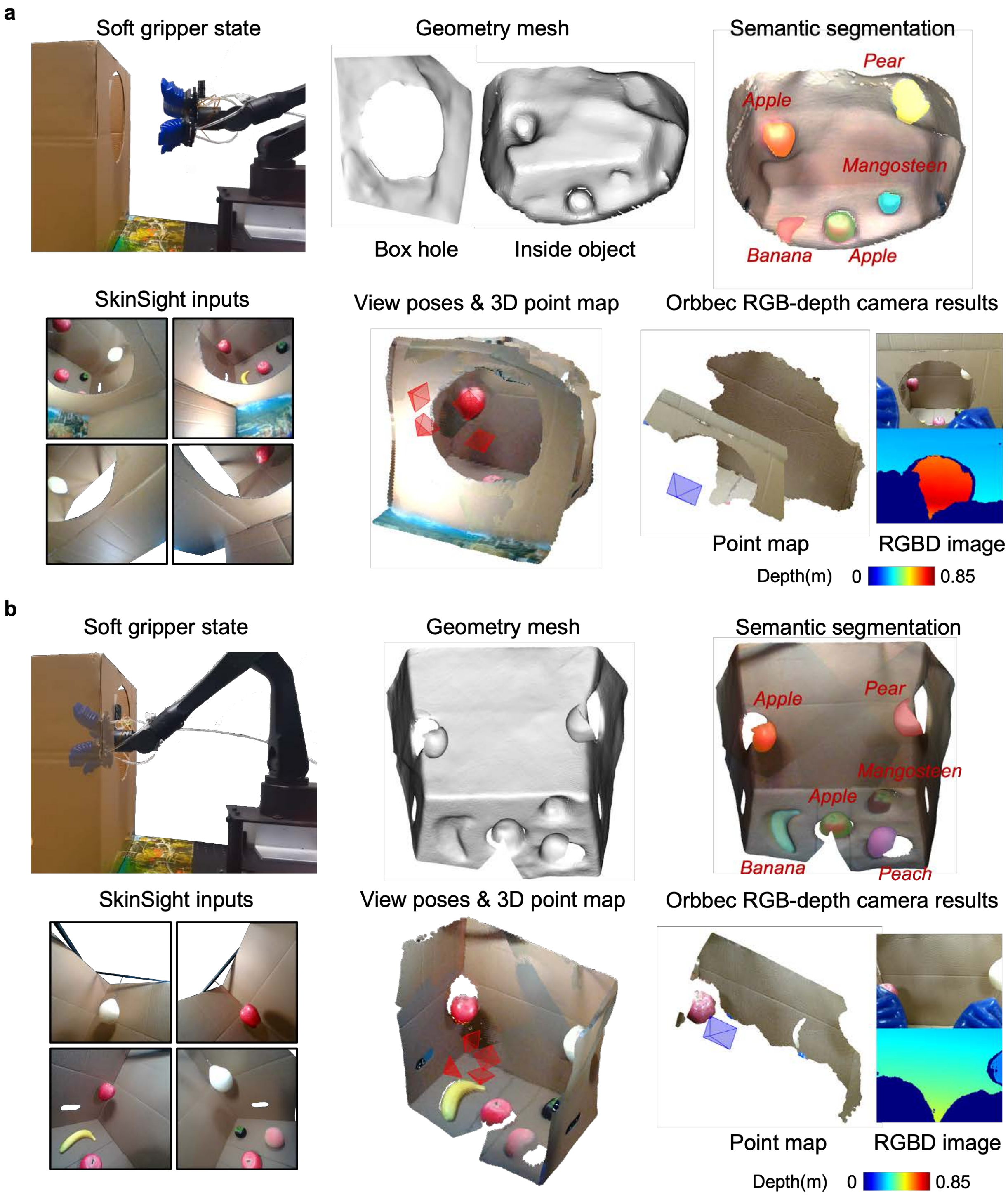

a
Soft gripper state
Geometry mesh
Semantic segmentation
Pear
Apple
Mangosteen
Box hole
Inside object
Banana
Apple
SkinSight inputs
View poses & 3D point map
Orbbec RGB-depth camera results
Point map
RGBD image
Depth(m) 0 0.85
b
Soft gripper state
Geometry mesh
Semantic segmentation
Apple
Pear
Mangosteen
Apple
Banana
Peach
SkinSight inputs
View poses & 3D point map
Orbbec RGB-depth camera results
Point map
RGBD image
Depth(m) 0 0.85

Fig. S11: Performance of SkinSight system on soft gripper. ab, Captured images and reconstruction results when soft gripper is **outside** and **inside** the box. The top row shows the state of the soft gripper, the reconstructed geometry mesh, and the semantic segmentation results. The bottom row illustrates the captured images by our SkinSight system, reconstruction view poses with 3D point map, and results from the Orbbec DaBai DCW2 RGB-depth camera for comparison. The results demonstrate that SkinSight successfully recovers the view poses and reconstructs the geometry of the box and the fruit objects inside. The rendered images are detailed enough to support visual tasks such as semantic segmentation. Compared to the conventional RGB-depth solution, SkinSight excels in flexibility, field of view (FoV), and robustness. Despite its lightweight design, SkinSight does not increase the system's volume or weight (Fig. 4ab). Its flexible FoV enables wide-range capturing and reconstruction of scenes both inside and outside the box. When the soft gripper is outside the box, SkinSight captures four complete fruit objects (two apples, one pear, and one mangosteen), along with half of a banana. Only the peach remains outside its FoV. When inside the box, SkinSight can fully capture and reconstruct all the fruit objects. In contrast, the RGB-depth camera has a much narrower FoV. Especially when positioned inside the box, it captures fewer objects due to the limited FoV. While its RGB camera can see the fruit, the depth camera requires strict working conditions for depth estimation. Under suboptimal conditions, this leads to incomplete or missing fruit objects in the point map, even when the RGB camera can see them.

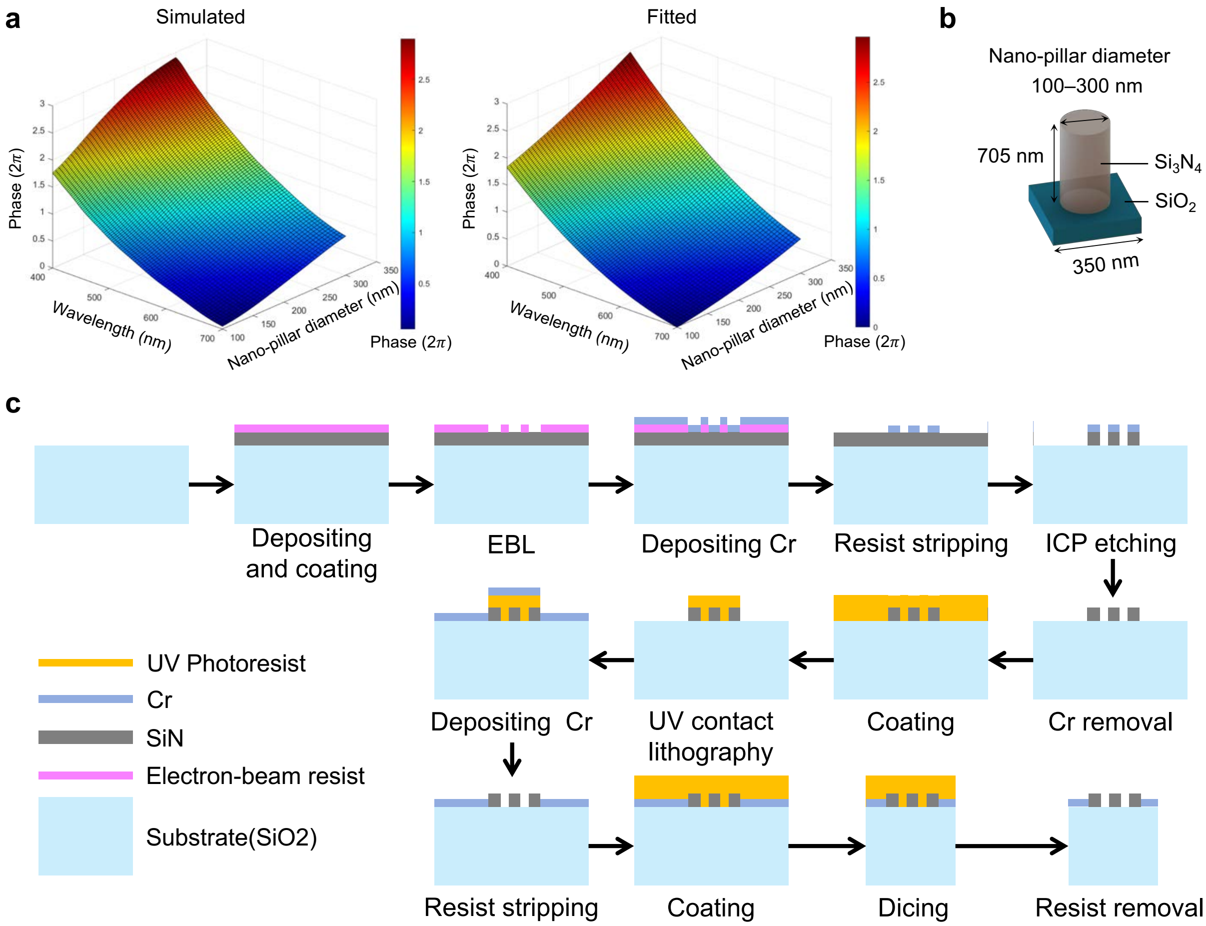


Fig. S12: Structure to phase mapping of nano-lens. a, Left, the simulated mapping from nano-pillar diameter and light wavelength to the modulation phase, computed using the finite-difference time-domain (FDTD) method (RCWA Solver in Lumerical, ANSYS Inc). Right, the fitted surface using polynomial $\varphi = 2\pi \times (6.051 - 0.0203\lambda + 2.26\eta + (1.371 \times 10^{-5})\lambda^2 - 0.00295\lambda\eta + 0.797\eta^2)$, where λ is the wavelength of the light in nano-meters, η and is the duty of the nano-pillar, and φ is the modulation phase. The duty is defined as: d_p/d_u, which d_p is the diameter of the nano-pillar, and $d_u = 350$ nm is the size of the square unit. **b,** The nano-pillar design. **c,** A schematic representation of the fabrication process for SiN-based devices on a SiO_2 substrate. The process begins with the deposition and coating of electron-beam resist, followed by electron beam lithography (EBL) to define the nanoscale pattern. Chromium (Cr) is deposited, and resist stripping is performed to create a Cr mask. This is followed by inductively coupled plasma (ICP) etching to transfer the pattern onto SiN nano-pillars. After the Cr mask is removed, UV photoresist is applied, and UV contact lithography is performed to define apertures for blocking stray light. The process continues with a second Cr deposition, resist stripping, and the application of an additional protective coating. Finally, the substrate is diced into 1×1 mm blocks, and the remaining resist is removed to complete the fabrication.

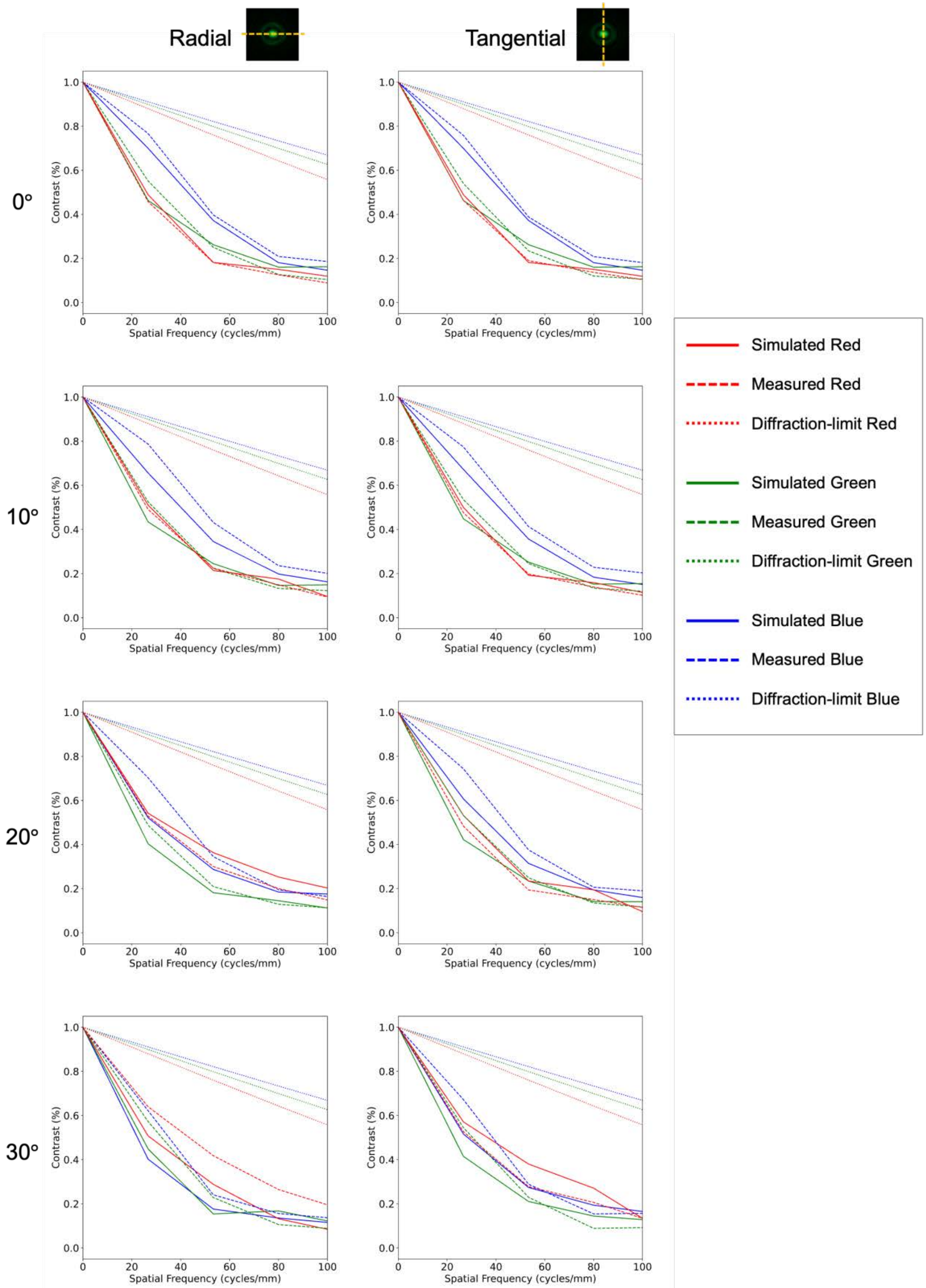


Fig. S13: The radial and tangential modulation transfer functions (MTFs) of the simulated and measured PSFs at incidence angles of 0°, 10°, 20°, and 30°. The diffraction-limit MTF curves are also plotted for comparison.

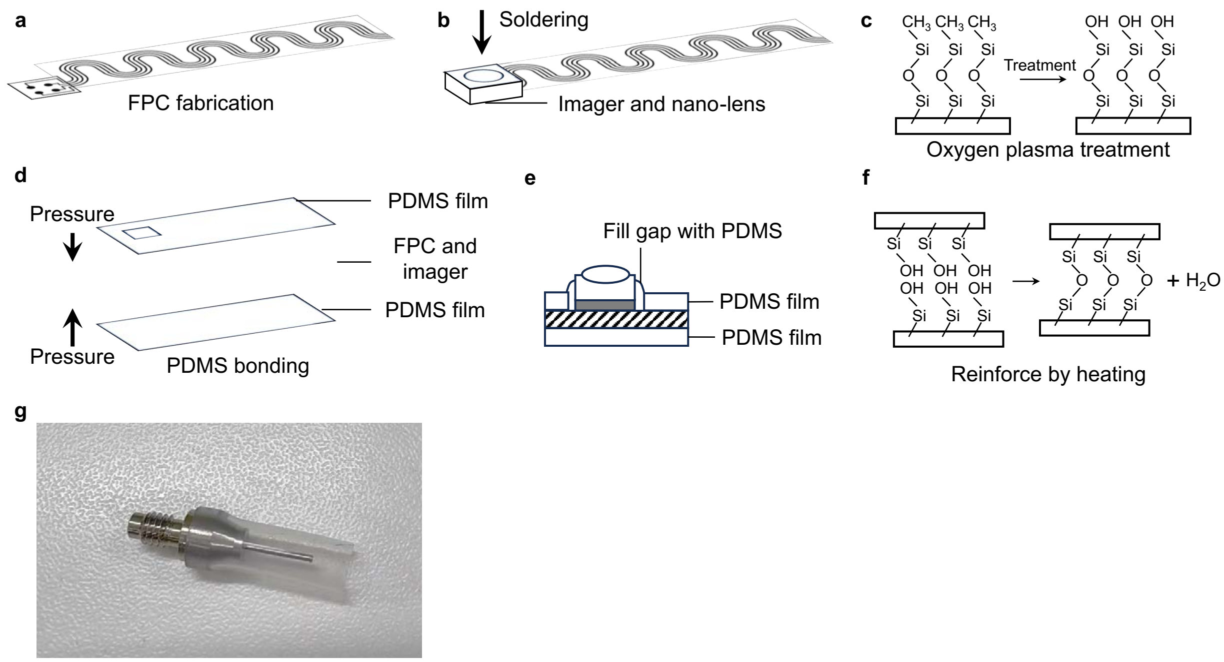


Fig. S14: Fabrication process of the SkinSight system. **a-f,** FPC preparation, imager soldering, oxygen plasma surface treatment, PDMS bonding, gap filling with PDMS, and heat curing for bond reinforcement. **g,** Custom vacuum nozzle for OV6946 sensor assembly.

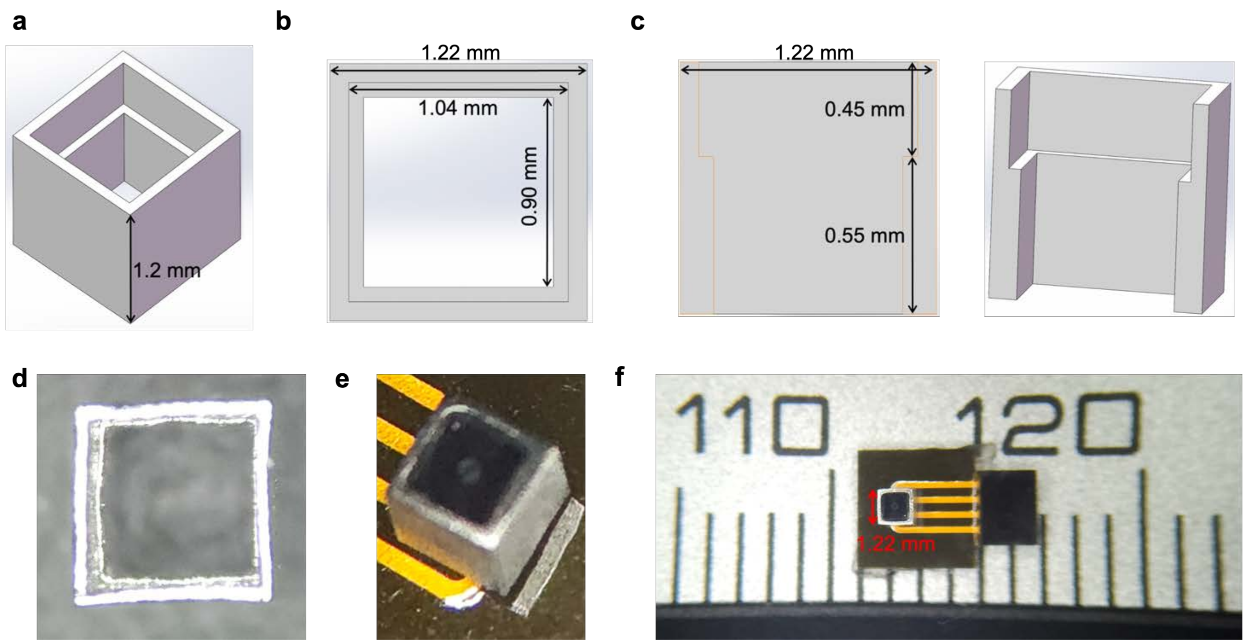


Fig. S15: 3D-printed metal shell for mounting the nano-lens and imager. The 3D-printed metal shell, with a printing accuracy of 10 μm (0.01 mm), is used to securely fix the nano-lens and imager. **a,** Global view of the metal shell. **b,** Top view of the shell. **c,** Two cross-sectional views of the shell. **d,** Optical image of the 3D-printed metal shell. **e,** Optical image showing the metal shell with the sensor and nano-lens assembled. **f,** Optical image depicting the placement of a sensor on a ruler, with the miniaturized camera size approximately 1 mm.

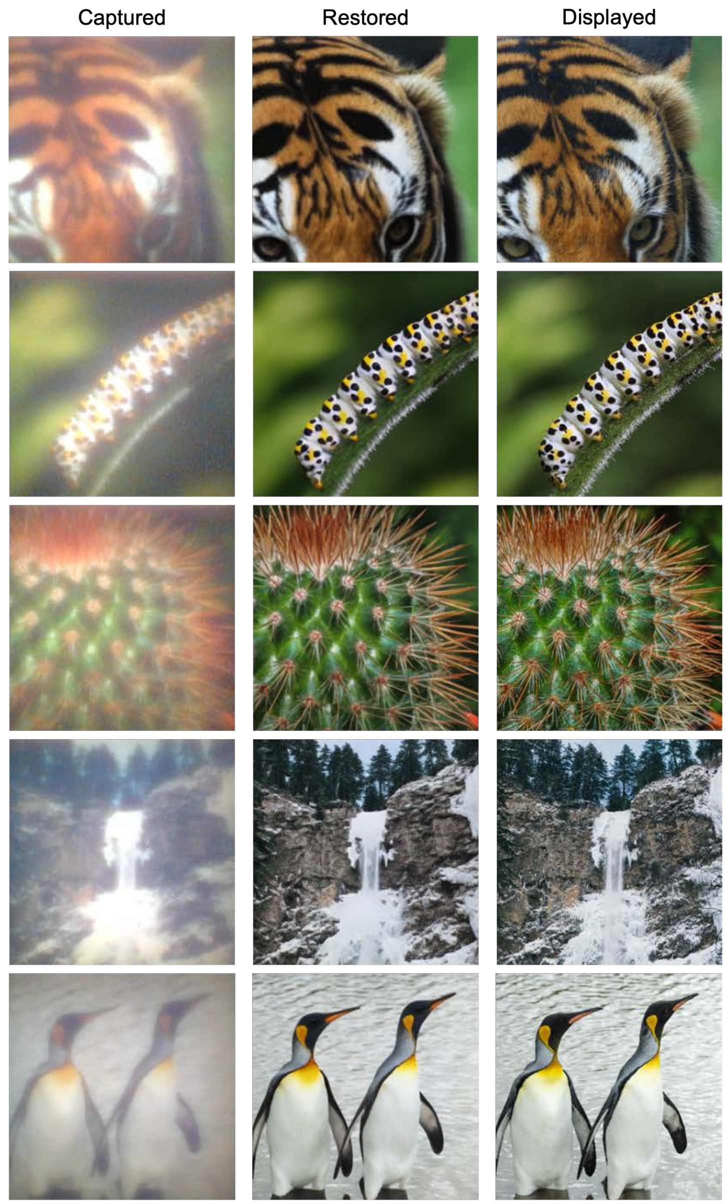


Fig. S16: Image restoration results.

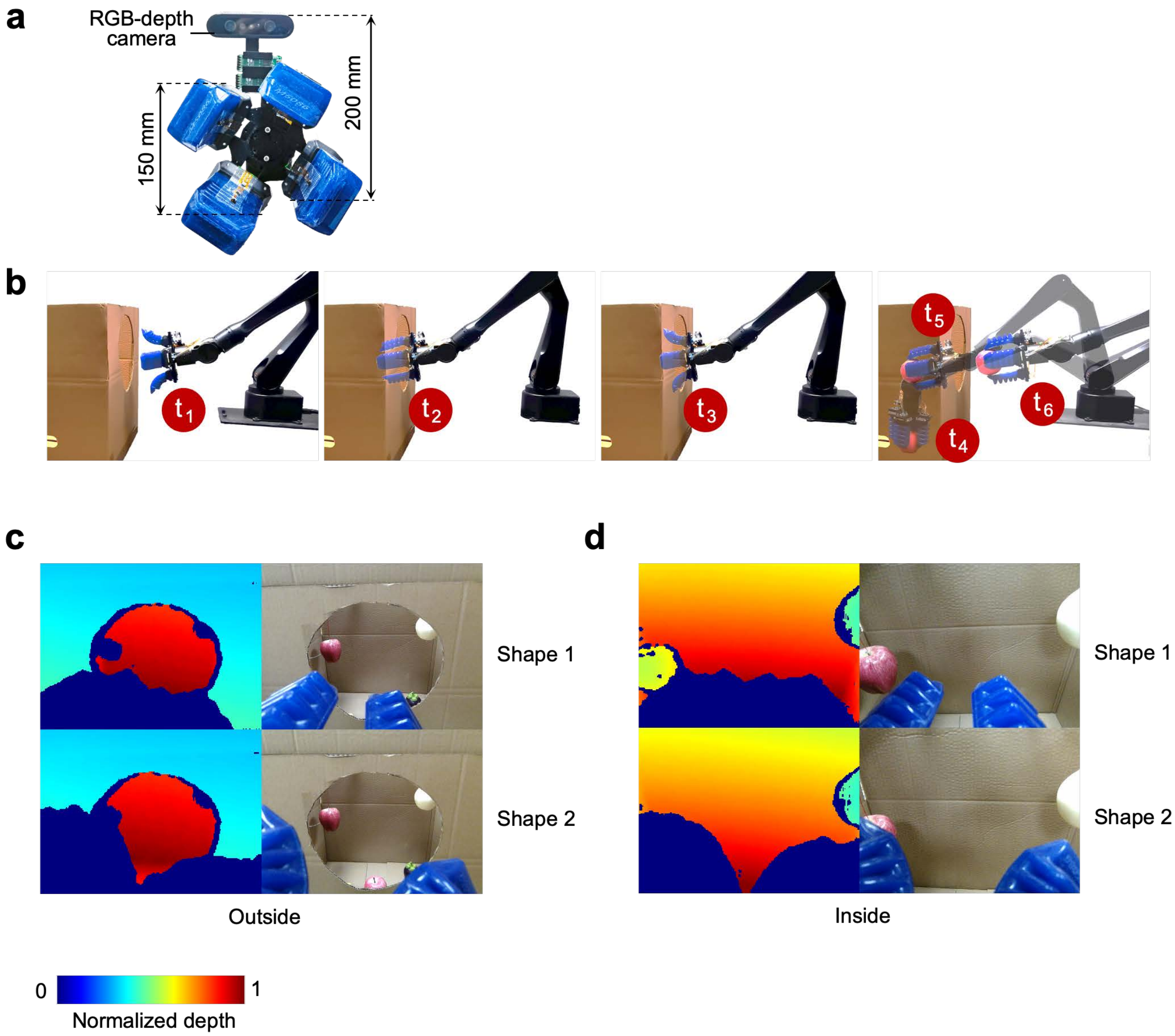


Fig. S17: An RGB-depth camera system for comparison. a, An RGB-depth camera (DaBai DCW2, Orbbec) is positioned beside the soft gripper for comparison. **b,** Six highlighted moments (t_1 to t_6) demonstrate how SkinSight helps the soft gripper in locating the target (t_1), passing through the narrow hole (t_2), reconstructing the objects (t_3), and grasping the object (t_4, t_5, t_6). **cd,** The captured depth and RGB images of the RGB-depth camera at position t_1 (outside the box) and t_3 (inside the box).

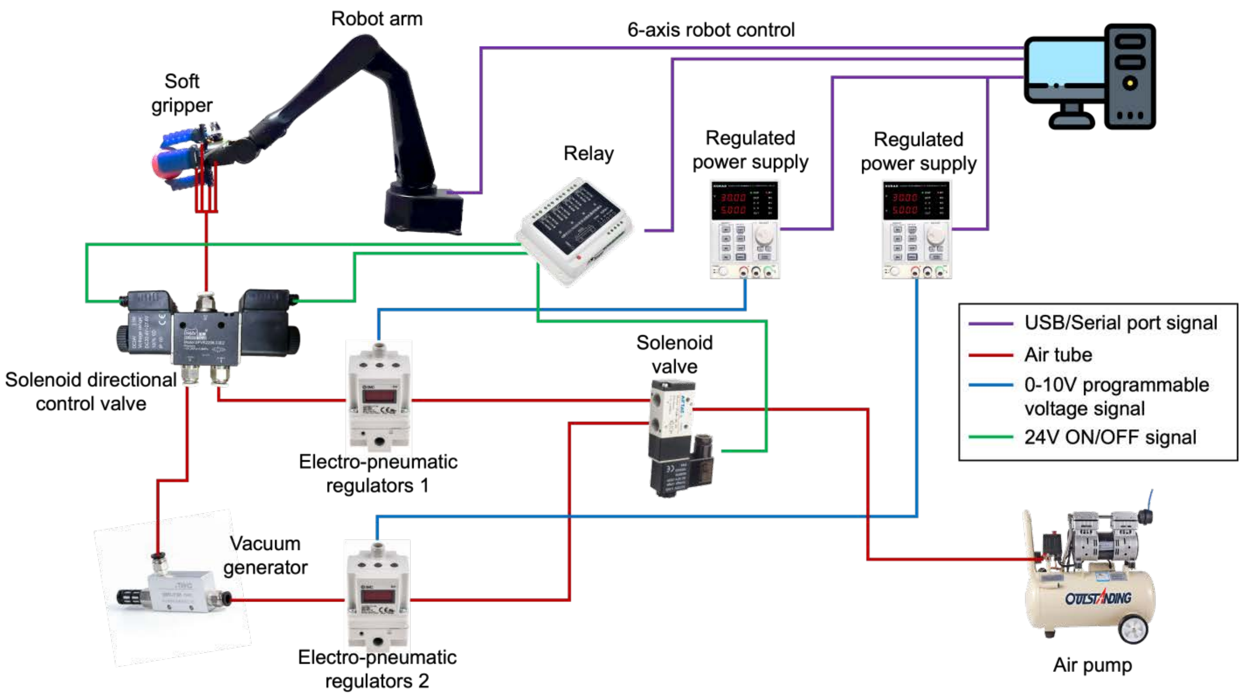


Fig. S18: Overview of the soft gripper system. The gripper is fixed on a 6-axis robotic arm (DISCOVER Robotics), which is controlled using the USB port of the PC. An air pump is used to provide air pressure. The pressure of the gripper is controlled by two SMC ITV1030 electro-pneumatic regulators, one for positive pressure and one for negative pressure. The negative pressure is generated using a vacuum generator. A solenoid directional control valve is used to switch the positive and negative pressure. The positive pressure range is 0~100 KPa, and the negative pressure range is -87~100 KPa. Two regulated power supplies are used to control the output pressure of the two electro-pneumatic regulators. The output voltage (0~10 V) of the egulated power supply is further programmed using the serial port of the PC. The ON/OFF state of the solenoid valves are controlled using a relay, which is also programmed using the serial port.

Supplementary Notes

Supplementary Note S1. AI-empowered deformation-adaptive reconstruction.

Asynchronous 3D reconstruction. To handle continuous topological deformations without pre-calibrated camera extrinsics, the spatial reconstruction is decoupled into an online geometric reconstruction pathway and an asynchronous Gaussian optimization pathway. The continuous multi-view video stream undergoes temporal chunking via a sliding window to yield overlapping frame chunks C_k. Each chunk $C_k = \{\boldsymbol{I}_1, \boldsymbol{I}_2, \ldots, I_L\}$ comprises a sequence of L images, which is processed as a collective input by the π3 visual geometry foundation model[1] $\Phi_{\pi 3}$. The functional mapping of the model is defined as:

$$\Phi_{\pi 3}(C_k) \rightarrow \{\boldsymbol{P}_t, \boldsymbol{c}_t, \boldsymbol{\Theta}_t\}_{t=1}^{L}$$

where $\boldsymbol{P}_t \in \mathbb{R}^{H \times W \times 3}$ represents the reconstructed dense 3D point map, $\boldsymbol{c}_t \in \mathbb{R}^{H \times W}$ denotes the per-pixel confidence score, and $\boldsymbol{\Theta}_t \in SE(3)$ signifies the locally consistent camera pose for each frame t within the chunk. By implicitly encoding cross-view semantic and geometric features, the model directly regresses these spatial representations in a local coordinate space without requiring explicit feature matching or camera calibration.

The relative transformation between adjacent chunks C_k and C_{k+1} can then be efficiently analytically solved using a least-squares formulation. Given the set of corresponding 3D points $\{\boldsymbol{P}_k\}$ and $\{\boldsymbol{P}_{k+1}\}$ derived from the overlapping point maps, we minimize the weighted registration error:

$$E = \sum c_k^i \, \rho(\|\boldsymbol{P}_k^i - (s\boldsymbol{R}\boldsymbol{P}_{k+1}^i + \boldsymbol{T})\|_2)$$

where $\boldsymbol{R}$, $\boldsymbol{T}$, and s represent the rotation matrix, translation vector, and scaling factor, respectively. c_k^i denotes the confidence value for the i-th point correspondence extracted from the reference chunk C_k. This approach ensures that the global coordinate alignment is primarily driven by high-confidence scene structures, with the confidence map effectively suppressing the influence of dynamic objects. Meanwhile, the Huber loss $\rho(\cdot)$ is utilized to handle the geometric outliers. This sequential alignment integrates all local chunks into a unified global coordinate system.

3D Gaussian representation. The 3D probabilistic representation approximates the scene by modeling it as a collection of 3D distributions[2–4]. Taking the Gaussian distribution as an example, each Gaussian distribution G_i is characterized by the following probability density function:

$$G_i(\mathbf{x}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) = \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^T\boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i)\right),$$

where $\boldsymbol{\mu}_i \in \mathbb{R}^3$ is the mean vector representing the position of the *i*th Gaussian distribution, and $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3\times3}$ is the covariance matrix that defines the Gaussian spread. Each Gaussian has 4 attributes, including the 3-dim color (red, green, blue) and opacity. The covariance matrix $\boldsymbol{\Sigma}_i$ can be factorized into a scaling matrix $\mathbf{S}_i = diag(S_1, S_2, S_3)$ and a rotation matrix $\mathbf{R}_i \in \mathbb{R}^{3\times3}$:

$$\boldsymbol{\Sigma}_i = \mathbf{R}_i\mathbf{S}_i\mathbf{S}_i^T\mathbf{R}_i^T,$$

where S_1, S_2, S_3 are the scaling factors along 3 axes, and the direction of the minimum scale factor corresponds to the normal direction $\mathbf{n}_i$ of the Gaussian. To further solve the ambiguity of the normal direction, we set the angle between the viewing direction (from camera viewpoint to the center of the Gaussian) and the normal direction $\mathbf{n}_i$ larger than 90 degrees.

With such a representation, the 2D images can be differentiably rendered using alpha blending. Assuming a camera with viewing transformation matrix **W** and intrinsic matrix **K**, *i*th 3D Gaussian distribution can be projected to the 2D image at pixel position u as $G_i(\mathbf{x}|\boldsymbol{\mu}_i', \boldsymbol{\Sigma}_i')$,

$$\mu_i' = \mathbf{KW}[\boldsymbol{\mu}_i, 1]^T,$$

$$\Sigma_i' = \mathbf{JW}\boldsymbol{\Sigma}_i\mathbf{W}^T\mathbf{J}^T.$$

where $\mathbf{J}$ is the Jacobian matrix of the affine approximation for the projective transformation from 3D space to the 2D image space. For alpha-blending rendering, all the Gaussian are sorted based on the depth, and contributes to the pixel color map **I**, distance map **D**, and normal map **N**:

$$\mathbf{I} = \sum_{i\in N} c_i\, a_i \prod_{j=1}^{i-1}(1-a_j),$$

$$\mathbf{D} = \sum_{i\in N} d_i\, a_i \prod_{j=1}^{i-1}(1-a_j),$$

$$d_i = \left(\mathbf{R}_c^T(\boldsymbol{\mu}_i - \mathbf{T}_c)\right)\mathbf{R}_c^T\mathbf{n}_i^T,$$

$$\mathbf{N} = \sum_{i\in N} \mathbf{R}_c^T\mathbf{n}_i\, a_i \prod_{j=1}^{i-1}(1-a_j).$$

Where c_i represents the opacity and color of the Gaussian G_i. c_i is computed by evaluating $G_i(u|\boldsymbol{\mu}_i', \boldsymbol{\Sigma}_i')$ multiplied by the Gaussian's opacity value, and u is the rendering pixel position . $\mathbf{R}_c$ is the rotation matrix from the image frame to the global coordinate system, $\mathbf{T}_c$ denotes the camera center or view point in world coordinate, μ_i is the Gaussian center. The depth map can be subsequently computed based on the distance map.

Gaussian reconstruction rendering. First, an initial 3D probabilistic representation can be generated based by converting each point in the 3D unified point map $\hat{\mathbf{X}}$ to a Gaussian distribution G_i. Then the mean (position), covariance (spread), color, and opacity of all the Gaussians are learned from the input images via gradient descent. The loss function is defined below, including the color, depth, and normal loss:

$$L = \lambda_c L_c + \lambda_D L_D + \lambda_N L_N,$$

$$L_c = \frac{1}{HW} \sum_{p=1}^{HW} \|\mathbf{I}_{in}(p) - \mathbf{I}(p)\|_2^2,$$

$$L_D = \frac{1}{HW} \sum_{p=1}^{HW} \|\mathbf{Dep}_{in}(p) - \mathbf{Dep}(p)\|_1,$$

$$L_N = \frac{1}{HW} \sum_{p=1}^{HW} |\nabla \mathbf{I}| \|\mathbf{N}_{in}(p) - \mathbf{N}(p)\|_1.$$

Where λ_C, λ_D, and λ_N are the weights for three different losses, $\mathbf{I}_{in}$ is the input images, $\mathbf{Dep}_{in}$ denotes the input distance map computed from the point map $\hat{\mathbf{X}}$ obtained in the image registration step and camera intrinsic matrix $\mathbf{K}$:

$$\mathbf{Dep}(p) = \mathbf{D}(p)/\mathbf{N}(p)\mathbf{K}^{-1}\tilde{p},$$

where p denotes the 2D position on the image, $\tilde{p}$ represents its homogeneous coordinate. $\mathbf{N}_{in}$ represents the normal map inferred from the input distance map $\mathbf{Dep}_{in}$. $\mathbf{I}$, $\mathbf{D}$, and $\mathbf{N}$ are the rendered color, distance, normal map from the 3D probabilistic representation, respectively.

We use gradient descent to optimize the position $\boldsymbol{\mu}_i$, covariance $\boldsymbol{\Sigma}_i$, color (red, green, blue), and opacity of all the Gaussians. After 1000 to 3000 iterations training, the color, distance/depth, and normal maps of a given camera (intrinsic matrix and extrinsic matrix) can be obtained by rendering. With rendered high-quality depth maps and registered camera poses, the geometry mesh can be reconstructed via the TSDF integration and marching cube algorithm.

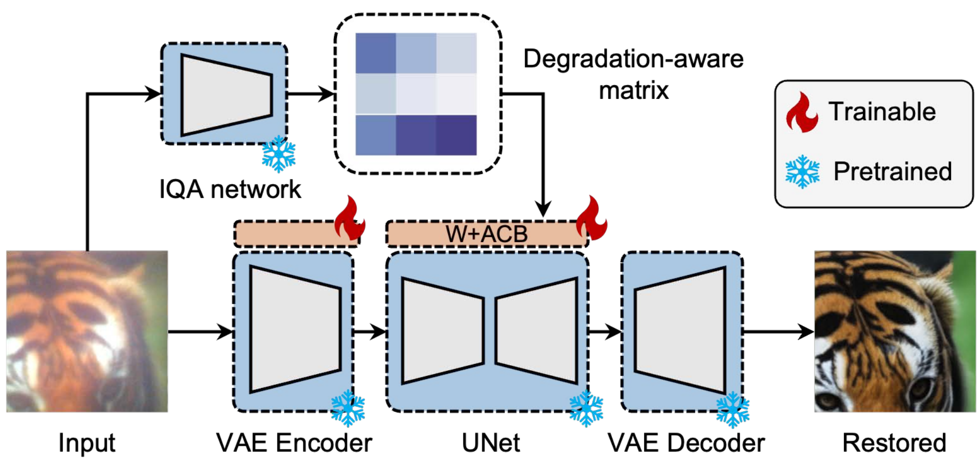
Supplementary Note S2. Image restoration algorithm.

Inference strategy and model selection: To address the intrinsic trade-off between computational efficiency and reconstruction fidelity, our framework adopts a modeling strategy aligned with specific performance objectives. For latency-sensitive tasks, a lightweight U-Net[5] architecture is employed to provide rapid inference for correcting optical aberrations and recovering clear images. Alternatively, for scenarios demanding maximum textural fidelity and perceptual quality, the restoration process leverages a diffusion-based generative network[6]. To adapt this large-scale model to the system-specific degradation, we utilize a Low-Rank Adaptation (LoRA) framework[7].

Low-Rank Adaptation (LoRA) framework: LoRA is a parameter-efficient fine-tuning method widely used for adapting large pre-trained models to specific tasks[7]. LoRA achieves this by freezing the majority of the model's original weights while introducing a low-rank decomposition for weight updates. Mathematically, the adjusted weights are expressed as:

$$\mathbf{W}' = \mathbf{W} + \Delta\mathbf{W}, \text{ where } \Delta\mathbf{W} = \mathbf{A} \cdot \mathbf{B}.$$

In this formulation, $\mathbf{A} \in \mathbb{R}^{d\times r}$ and $\mathbf{B} \in \mathbb{R}^{r\times d}$ $(r \ll d)$ are trainable low-rank matrices. This decomposition significantly reduces the number of trainable parameters, making fine-tuning computationally efficient while preserving the base model's performance. For image restoration tasks, LoRA is typically applied to specific layers of the U-Net in Stable Diffusion[8] (SD-turbo checkpoint), enabling efficient adaptation to new domains with minimal overhead.



Degradation-aware LoRA: While the standard LoRA framework demonstrates strong performance, it does not explicitly account for the degradation characteristics of the input images. Here, the aberration of the nano-lens and the noise of the photodetectors. To address this limitation, we propose an enhancement by introducing a degradation-aware modulation matrix **C** into the LoRA framework[9]. In our method, the weight update is modified as:

$$\mathbf{W}' = \mathbf{W} + \Delta\mathbf{W}, \text{ where } \Delta\mathbf{W} = \mathbf{A} \cdot \mathbf{C} \cdot \mathbf{B}.$$

Here, $\mathbf{A} \in \mathbb{R}^{d\times r}$ and $\mathbf{B} \in \mathbb{R}^{r\times d}$ $(r \ll d)$ are the low-rank matrices as in the standard LoRA framework, while $\mathbf{C} \in \mathbb{R}^{r\times r}$ is a degradation-aware matrix that dynamically adjusts the weight update based on the specific quality characteristics of the input image. To compute the degradation-aware matrix **C**, we first perform image quality assessment (IQA) on the input images. A sliding window approach is used to extract patch-wise degradation metrics, which are computed using the MUSIQ[10] model. This process results in a degradation map **D** that captures spatial quality variations. The degradation map **D** is then processed through a Multi-Layer Perceptron (MLP) to produce the degradation-aware matrix **C**. This matrix reflects the degradation characteristics of the input image and dynamically modulates the low-rank updates to better align with the restoration task. The introduction of the degradation-aware matrix **C** enables the model to adapt its updates dynamically based on input-specific quality characteristics. This enhancement improves the restoration quality while maintaining the computational and storage efficiency of the original LoRA framework. The proposed method effectively leverages task-specific information to achieve high-quality and realistic image restoration results. To optimize the model for image restoration, we employ a combination of Mean Squared Error (MSE) and Learned Perceptual Image Patch Similarity (LPIPS) as the loss function[11]. The MSE loss ensures pixel-wise accuracy by minimizing the squared differences between the restored and ground truth images, while the LPIPS loss emphasizes perceptual similarity by aligning feature representations in a pre-trained neural network. This dual-loss formulation balances low-level accuracy and high-level perceptual quality, resulting in realistic and visually consistent outputs.

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